Foundational

$$P(E_1) + P(E_2) + \cdots = P(E_1 \cup E_2 \cup \cdots)$$

$$\mathbb{E}[X] = \sum_x x\,\mathbb{P}(X = x)$$

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\,dx$$

$$\mathrm{Var}(X) = \mathbb{E}\Big[(X - \mathbb{E}[X])^2\Big]$$

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

$$\mathbb{E}[X] = \sum_x x p(x)$$

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

$$\mathbb{E}[aX + b] = a\,\mathbb{E}[X] + b$$

$$\mathrm{Var}(aX + b) = a^2 \mathrm{Var}(X)$$

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$$

$$\mathrm{Cov}(X, Y) = \rho_{X,Y}\,\sigma_X\,\sigma_Y$$

Finance

$$\text{HPR} = \frac{\text{Price at the end of HP} - \text{Purchase Price} + \text{Dividends}}{\text{Purchase Price}}$$

Gross Return:

$$R_{t+1} = \frac{P_t - P_s}{P_s} + 1 = \frac{P_t}{P_s}$$

Log Return:

$$\log \text{return}_t = \log(R_t + 1) = \log\left(\frac{P_t}{P_s}\right)$$

If $\log \text{return}_t$ is sufficiently small:

$$\log \text{return}_t \approx R_t$$

$$\mathbb{E}[r_f] = r_f \qquad \mathrm{Var}(r_f) = 0$$

$\Rightarrow r_f$ is treated as a constant.

$$\text{Risk Premium} = \mathbb{E}[R] - r_f$$

$$\text{Excess Returns} = R - r_f$$

$$\text{Sharpe Ratio} = \frac{\text{Risk Premium}}{\text{Standard Deviation}} = \frac{\mathbb{E}[R] - r_f}{\mathrm{std}(R)}$$

Portfolio

$$U(R) = \mathbb{E}[R] - \frac{1}{2}A\sigma^2 = r_f + \theta\left(\mathbb{E}[R] - r_f\right) - \frac{1}{2}A(\theta^2\sigma^2)$$

$$\theta^* = \frac{\mathbb{E}[R] - r_f}{A\sigma^2}$$

Capital Allocation Line:

$$\mathbb{E}[R_p] = r_f + \text{Sharpe Ratio} \times \sigma_p$$

Two-Asset Portfolio Variance:

$$\sigma_p^2 = w_d^2\sigma_d^2 + w_e^2\sigma_e^2 + 2w_d w_e \sigma_d \sigma_e \rho_{de}$$

Case 1: when $\rho_{de} = 1$

$$\sigma_p = w_d\sigma_d + w_e\sigma_e$$

Case 2: when $\rho_{de} = -1$

$$\sigma_p = |w_d\sigma_d - w_e\sigma_e|$$

when $\sigma_p = 0$:

$$w_d = \frac{\sigma_e}{\sigma_d + \sigma_e}, \qquad w_e = \frac{\sigma_d}{\sigma_d + \sigma_e}$$

Case 3: when $-1 < \rho_{de} < 1$

$$\sigma_p = \sqrt{w_d^2\sigma_d^2 + w_e^2\sigma_e^2 + 2w_d w_e \sigma_d \sigma_e \rho_{de}}$$

when σ_p^2 is minimized:

$$w_d = \frac{\sigma_e^2 - \mathrm{Cov}(R_d, R_e)}{\sigma_d^2 + \sigma_e^2 - 2\mathrm{Cov}(R_d, R_e)} \quad \text{and} \quad w_e = 1 - w_d$$

Assuming all assets share similar variance and covariance

$$\sigma_p^2 = \frac{1}{N}\sigma^2 + \frac{N-1}{N}\sigma^2\rho$$

As $N \to \infty$:

$$\lim_{N\to\infty} \sigma_p^2 = \sigma^2\rho$$

Return Predictability

Random Walk 1:

$$r_t = \mu + \varepsilon_t, \quad \varepsilon_t \sim i.i.d.\,(0, \sigma^2)$$

$$p_t = p_{t-1} + \mu + \varepsilon_t, \quad \varepsilon_t \sim i.i.d.\,(0, \sigma^2)$$

$$\mathbb{E}(p_t \mid p_0) = p_0 + \mu t$$

$$\mathrm{Var}(p_t \mid p_0) = \sigma^2 t$$

Random Walk 2:

$$\mathbb{E}(r_t \mid \mathcal{F}_{t-1}) = \mu$$

Random Walk 3:

$$\mathbb{E}(r_t) = \mathbb{E}(r_{t+\tau}) = \mu \quad \text{and} \quad \mathrm{Cov}(r_t, r_{t+\tau}) = 0 \quad (\tau > 0)$$

Volatility

Realized variance (RV) and sample variance of log-returns:

$$\hat{\sigma}^2_{\mathrm{RV}} := \sum_{j=1}^{T} (r_t - r_{t-1})^2$$

$$\sigma^2 := \frac{1}{T-1}\sum_{j=1}^{T} (r_t - \bar{r})^2$$

where r_t is the log-return at time t:

$$r_t = \log P_t - \log P_{t-1} = \log\left(\frac{P_t}{P_{t-1}}\right)$$

Alternatively, one can use simple (percentage) returns:

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} \quad (\text{instead of } r_t)$$

Conditional Volatility:

$$\mathrm{Var}\left(R_t \;\middle|\; \mathcal{F}_{t-1}\right)$$

ARCH model:

$$r_t = \sigma_t^2\epsilon_t^2, \qquad \sigma_t^2 = \beta_0 + \sum_{i=1}^{q}\beta_i r_{t-i}^2,$$

GARCH model:

$$r_t^2 = \epsilon_t^2\left(\beta_0 + \sum_{i=1}^{q}\beta_i r_{t-i}^2 + \sum_{j=1}^{p}\gamma_j\sigma_{t-j}^2\right)$$

Linear Time Series

Weak Stationarity: if

$$\mathbb{E}(X_t) \text{ and } \mathrm{Var}(X_t) \text{ are fixed for all } t$$

$$\mathrm{Cov}(X_t, X_{t-\tau}) \text{ does not depend on } t$$

Strict Stationarity: if the distribution of $(X_1, X_2, \ldots, X_n)$ is the same as that of $(X_{1+k}, X_{2+k}, \ldots, X_{n+k})$ for all n and k
Strict Stationarity Preservation: If X_t strictly stationary and

$$Z_t = f(X_t, X_{t-1}, X_{t-2}, \ldots)$$

for continuous f, then Z_t strictly stationary. Wold's Theorem (Weakly Stationary Process):

$$X_t = \mu + \sum_{j=0}^{\infty} b_j\varepsilon_{t-j}, \quad \sum_{j=0}^{\infty} b_j^2 < \infty$$

ε_t: mean-zero white noise
A time series ε_t is a white noise with mean μ and variance σ^2 if

- $\mathbb{E}(\varepsilon_t) = \mu$ for all t,
- $\mathrm{Var}(\varepsilon_t) = \sigma^2 > 0$ for all t,
- $\mathrm{Cov}(\varepsilon_t, \varepsilon_s) = 0$ for all $t \neq s$.

Autocovariance (at lag τ), equal to variance at lag 0:

$$\gamma(\tau) = \mathrm{Cov}(X_t, X_{t-\tau}) = \mathbb{E}\left[(X_t - \mu)(X_{t-\tau} - \mu)\right]$$

Law of Large Numbers for Time Series: If X_t stationary and $\sum_{s=-\infty}^{\infty}|\gamma(s)| < \infty$,

$$\bar{X}_T = \frac{1}{T}\sum_{t=1}^{T} X_t \xrightarrow{p} \mu$$

Ergodicity Relaxation:

$$\lim_{T\to\infty}\frac{1}{T}\sum_{s=1}^{T}|\gamma(s)| = 0$$

(finite second moment) Ergodic Theorem. If a time series X_t is stationary and ergodic, then

$$\frac{1}{T}\sum_{t=1}^{T} X_t \xrightarrow{p} \mu$$

where $\mu = \mathbb{E}X_t < \infty$.
Autocorrelation Function (ACF):

$$\rho(\tau) = \mathrm{Corr}(X_t, X_{t+\tau})$$

Sample ACF:

$$\hat{\rho}(\tau) = \frac{\hat{\gamma}(\tau)}{\hat{\gamma}(0)}, \quad \hat{\gamma}(\tau) = \frac{1}{T}\sum_{t=1}^{T}(X_{t+\tau} - \bar{X})(X_t - \bar{X})$$

Conditional Expectation Basics:

$$\mathbb{E}(X_t \mid X_{t-1}, X_{t-2}, \ldots) = E_{t-1}(X_t)$$

or

$$\mathbb{E}(X_t \mid \mathcal{F}_{t-1}), \quad \mathcal{F}_{t-1} = \sigma(X_{t-1}, X_{t-2}, \ldots) \text{ (natural filtration)}$$

Adapted Process: X_t adapted to $\mathcal{G}_t$ if $\mathbb{E}(X_t \mid \mathcal{G}_t) = X_t$
Martingale Difference Sequnece:

$$E[e_t \mid \mathcal{F}_{t-1}] = 0$$

Unforecastability:

$$E(X_t \mid \mathcal{F}_{t-1}) = E(X_t)$$

If a time series X_t is stationary and ergodic martingale difference with $EX_t^2 = \sigma^2 < \infty$, then as $T \to \infty$ we have:

$$\frac{1}{\sqrt{T}}\sum_{t=1}^{T} X_t \xrightarrow{d} N(0, \sigma^2) \tag{1}$$

$$\mathrm{Var}\left(\frac{1}{\sqrt{T}}\sum_{t=1}^{T} X_t\right) = \sum_{u=-(T-1)}^{T-1}\left(1 - \frac{|u|}{T}\right)\gamma(u).$$

$$\lim_{T\to\infty}\frac{1}{T}\sum_{u=0}^{T-1}\left(1 - \frac{u}{T}\right)\gamma(u) = \sum_{j=0}^{\infty}\gamma(j)$$

If X_t satisfies mixing conditions, we have $\sum_{j=0}^{\infty}\gamma(j) \leq C < \infty$,

in which case

$$S_n = \frac{1}{\sqrt{T}}\sum_{t=1}^{T} X_t \xrightarrow{d} N\left(0, \sum_{j=-\infty}^{\infty}\gamma(j)\right).$$

AR Process:

$$X_t = \sum_{j=1}^{p}\phi_j X_{t-j} + \varepsilon_t$$

AR(1) Process

$$X_t = \phi_1 X_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \mathrm{WN}(0, \sigma_\varepsilon^2)$$

Infinite MA representation (stationary case):

$$X_t = \sum_{k=0}^{\infty}\phi_1^k\varepsilon_{t-k}$$

Mean: $\mathbb{E}(X_t) = 0$
Variance:

$$\mathrm{Var}(X_t) = \sigma_\varepsilon^2\sum_{k=0}^{\infty}\phi_1^{2k}$$

Converges iff $|\phi_1| < 1$:

$$\mathrm{Var}(X_t) = \frac{\sigma_\varepsilon^2}{1 - \phi_1^2} \quad \text{for } |\phi_1| < 1$$

Factor Model and PCA

$$\mathbf{r}_t = \alpha + \beta\mathbf{f}_t + \varepsilon_t$$

where $\mathbf{r}_t$, α and ε_t are $k \times 1$, β is $k \times m$, and $\mathbf{f}_t$ is $m \times 1$.

High Frequency Financial Econometrics

Geometric Brownian Motion:

$$dX_t = \mu X_t dt + \sigma X_t dW_t$$

Derivatives

Call Option:

$$c = P_0 N(d_1) - Ke^{-rT}N(d_2)$$

$$d_1 = \frac{\ln(P_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$